1.punkt

Linux: /etc/shadow polecenie dla obejrzania: sudo cat /etc/shadow postać: imie:$algorytm$sol$hash.

Windows: SAM-plik (C:\Windows\System32\config\SAM), on jest zablokowany systemem. Można go dostać tylko oprgramowaniem spiecjalnym.

2.punkt

Zawartosc pliku przykladowytekst.txt:

```
Cryptography or cryptology is the practice and study of techniques for secure
communication in the presence of adversarial behavior More generally cryptography is
about constructing and analyzing protocols that prevent third parties or the public
from reading private messages Modern cryptography exists at the intersection of the
disciplines of mathematics computer science information security electrical
engineering digital signal processing physics and others
```

Skroty dp slow dziendobry i milosc:

```
C:\Users\student>echo | set /p="dziendobry" > %TMP%\hash.txt

C:\Users\student>type %TMP%\hash.txt
dziendobry
C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA256 | findstr /v "hash"
25ae7545c10e93b1b450bcbbc91ec2355adb125b021ccade121414c4e17a5295

C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA1 | findstr /v "hash"
c2f3ce978fcfcdae53d71f64ad0387cce20be0a3

C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA384 | findstr /v "hash"
353c3fe5954aead481c088a8f943a4617bed96d3d2f4c2352639c69b8440433db198a30c82b3e7909ecaf8802aa19444

C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA512 | findstr /v "hash"
5a6d80a6fb6f82d441cc7b4e95e8071ae3a5b1d1a81ed46ef0d6b4dadb5e424f046859da957ae75a6bd22d2a2265c15ce6215ace44958e43ba7d03ac90fb76fa

C:\Users\student>certutil -hashfile %TMP%\hash.txt md5 | findstr /v "hash"
7eaae73ccda9de7b938dd567d145bf1e

C:\Users\student>echo | set /p="milosc" > %TMP%\hash.txt

C:\Users\student>type %TMP%\hash.txt
milosc
C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA1 | findstr /v "hash"
a6e6c518d737471a08d3fa94f5fa7ed7873c22e0

C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA256 | findstr /v "hash"
f318f3638c5678ae4d6aa07729a7c0a2c7447002e8938fe8ed9688a11ce0883e

C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA384 | findstr /v "hash"
00240c9dc33d72730d3578039aaf167d217c42dc6d95e5d5fad94528b8c15116b000713dc9db22250814b2d4ece04001

C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA512 | findstr /v "hash"
96906a57ada11ccd4abd7958c5f574ae5b7fffb3cda227048cce5fdd474f9069b76fffd82a97d8657f4cdc388892f149282ab472de5ae9db2997e7fc41aae66c

C:\Users\student>certutil -hashfile %TMP%\hash.txt md5 | findstr /v "hash"
0503da584484323f6491dbc15eae465f

C:\Users\student>
```

Skroty dla plika:

```
C:\Users\student>certutil -hashfile "przykladowytekst.txt" SHA1
SHA1 hash of przykladowytekst.txt:
6cbfcdde29d8c2ff186a37e525e1b190d3b42a79
CertUtil: -hashfile command completed successfully.

C:\Users\student>certutil -hashfile "przykladowytekst.txt" SHA256
SHA256 hash of przykladowytekst.txt:
9a16643b668247c5cab2271b894fa3210bf4fb42ad154736d36e52fc9e3cfc5f
CertUtil: -hashfile command completed successfully.

C:\Users\student>certutil -hashfile "przykladowytekst.txt" SHA384
SHA384 hash of przykladowytekst.txt:
e89d9e05ae5972974aadf27ab91f3004ee43e978c3b468b9702c184f8152b766f05c1637010921ca410c4c3caf73c79e
CertUtil: -hashfile command completed successfully.

C:\Users\student>certutil -hashfile "przykladowytekst.txt" SHA512
SHA512 hash of przykladowytekst.txt:
629cc0f309490baaba021faa087813fbf1f06d41adf94ceacf40a7c8d9c3a2b1cecb93578d47f46bc3ca53e06d3f8f25eb6f2928984b9ae34e11ccc0531d43b9
CertUtil: -hashfile command completed successfully.

C:\Users\student>certutil -hashfile "przykladowytekst.txt" md5
MD5 hash of przykladowytekst.txt:
62268896d1250eca0d842055f2ecacba
CertUtil: -hashfile command completed successfully.

C:\Users\student>
```

slowo/funkcja sha1 sha2-256 sha2-384 sha2-512 md5

dziendobry 160 bit=20 bajt 256bit=32 bajt 384 bit=48 bajt 512 bit=64 bajt 128 bit=16 bajt
milosc 160 bit=20 bajt 256bit=32 bajt 384 bit=48 bajt 512 bit=64 bajt 128 bit=16 bajt

plik: 160 bit=20 bajt 256bit=32 bajt 384 bit=48 bajt 512 bit=64 bajt 128 bit=16 bajt

slowo/funkcja	SHA1	SHA2-256	SHA2-384	SHA2-512	MD5
dziendobry	160 bit/20bajt	256 bit/32 bajt	384 bit/48 bajt	512 bit/64 baj	128 bit/16 bajt
milosc	160 bit/20bajt	256 bit/32 bajt	384 bit/48 bajt	512 bit/64 baj	128 bit/16 bajt
plik	160 bit/20bajt	256 bit/32 bajt	384 bit/48 bajt	512 bit/64 baj	128 bit/16 bajt

3. punkt

Porownanie skrotow

```
sha1
plik            6cbfcdde29d8c2ff186a37e525e1b190d3b42a79
dziendobry      c2f3ce978fcfcdae53d71f64ad0387cce20be0a3
milosc          a6e6c518d737471a08d3fa94f5fa7ed7873c22e0

sha2-256
plik            9a16643b668247c5cab2271b894fa3210bf4fb42ad154736d36e52fc9e3cfc5f
dziendobry      25ae7545c10e93b1b450bcbbc91ec2355adb125b021ccade121414c4e17a5295
milosc          f318f3638c5678ae4d6aa07729a7c0a2c7447002e8938fe8ed9688a11ce0883e

sha512
plik            629cc0f309490baaba021faa087813fbf1f06d41adf94ceacf40a7c8d9c3a2b1cecb93578d47f46bc3ca53e06d3f8f25eb6f2928984b9ae34e11ccc0531d43b9
dziendobry      5a6d80a6fb6f82d441cc7b4e95e8071ae3a5b1d1a81ed46ef0d6b4dadb5e424f046859da957ae75a6bd22d2a2265c15ce6215ace44958e43ba7d03ac90fb76fa
milosc          96906a57ada11ccd4abd7958c5f574ae5b7fffb3cda227048cce5fdd474f9069b76fffd82a97d8657f4cdc388892f149282ab472de5ae9db2997e7fc41aae66c

md5
plik            62268896d1250eca0d842055f2ecacba
dziendobry      7eaae73ccda9de7b938dd567d145bf1e
milosc          0503da584484323f6491dbc15eae465f
```

4. punkt

Wybrałem słowo milosc

```
C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA256 | findstr /v "hash"
f318f3638c5678ae4d6aa07729a7c0a2c7447002e8938fe8ed9688a11ce0883e
```

```
┌──(kali㉿kali)-[~]
└─$ echo -n "milosc" | sha256sum
f318f3638c5678ae4d6aa07729a7c0a2c7447002e8938fe8ed9688a11ce0883e  -
```

Wniosek: skroty mają tą samą długość i wartość.

5. punkt

Porownanie skrotow "milosc" i "milasc"

```
C:\Users\student>type %TMP%\hash.txt
milosc
C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA256 | findstr /v "hash"
f318f3638c5678ae4d6aa07729a7c0a2c7447002e8938fe8ed9688a11ce0883e

C:\Users\student>echo | set /p="milasc" > %TMP%\hash.txt

C:\Users\student>type %TMP%\hash.txt
milasc
C:\Users\student>certutil -hashfile %TMP%\hash.txt SHA256 | findstr /v "hash"
2bcf9272bce1ac19f24ff5f2c649a24892aecf353d1323d2c8d49c662ad1fbe5

C:\Users\student>
```